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Case 32-2025: A 79-Year-Old Man with Dyspnea, Edema, and Pacemaker-Lead Displacement

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PRESENTATION OF CASE

Dr. Zhuoshan Huang: A 79-year-old man was admitted to the Third Affiliated Hospital of Sun Yat-sen University in Guangzhou, China, because of dyspnea and edema in the arms and legs.

The patient had been in his usual state of health until 17 months before the current admission, when he was admitted to another hospital because of an acute inferolateral ST-elevation myocardial infarction for which he received a drug-eluting stent in the left circumflex artery. During that admission, computed tomography (CT) of the chest, performed without the administration of intravenous contrast material, reportedly showed coronary-artery calcification but was otherwise normal. Treatment with aspirin, clopidogrel, atorvastatin, perindopril, and metoprolol was started, and the patient was discharged home. Three months later, he underwent planned staged percutaneous coronary intervention for a chronic occlusion of the right coronary artery.

Two months before the current admission, the patient presented to the Third Affiliated Hospital of Sun Yat-sen University because of exertional chest discomfort. An electrocardiogram (Fig. 1A) showed sinus arrest with junctional bradycardia and retrograde P waves. The blood level of thyrotropin was normal. Treatment with metoprolol was stopped. The results of transthoracic echocardiography were reportedly normal. Coronary angiography showed patent stents in the left circumflex and right coronary arteries. A permanent dual-chamber pacemaker was implanted, but there was procedural difficulty during implantation of the atrial lead, with low sensing and high pacing thresholds obtained at multiple sites in the right atrium where implantation attempts were made, including the right atrial appendage. Ultimately, an unconventional implantation site (the ostium of the right atrial appendage) was chosen, which resulted in a pacing threshold of 0.5 V (reference value, <1.0), a sensing threshold of 1.5 mV (reference value, ≥1.5), and a pacing impedance of 722 ohms (reference range, 400 to 1000) in the right atrial lead.

Author affiliations are listed at the end of the article.

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CME



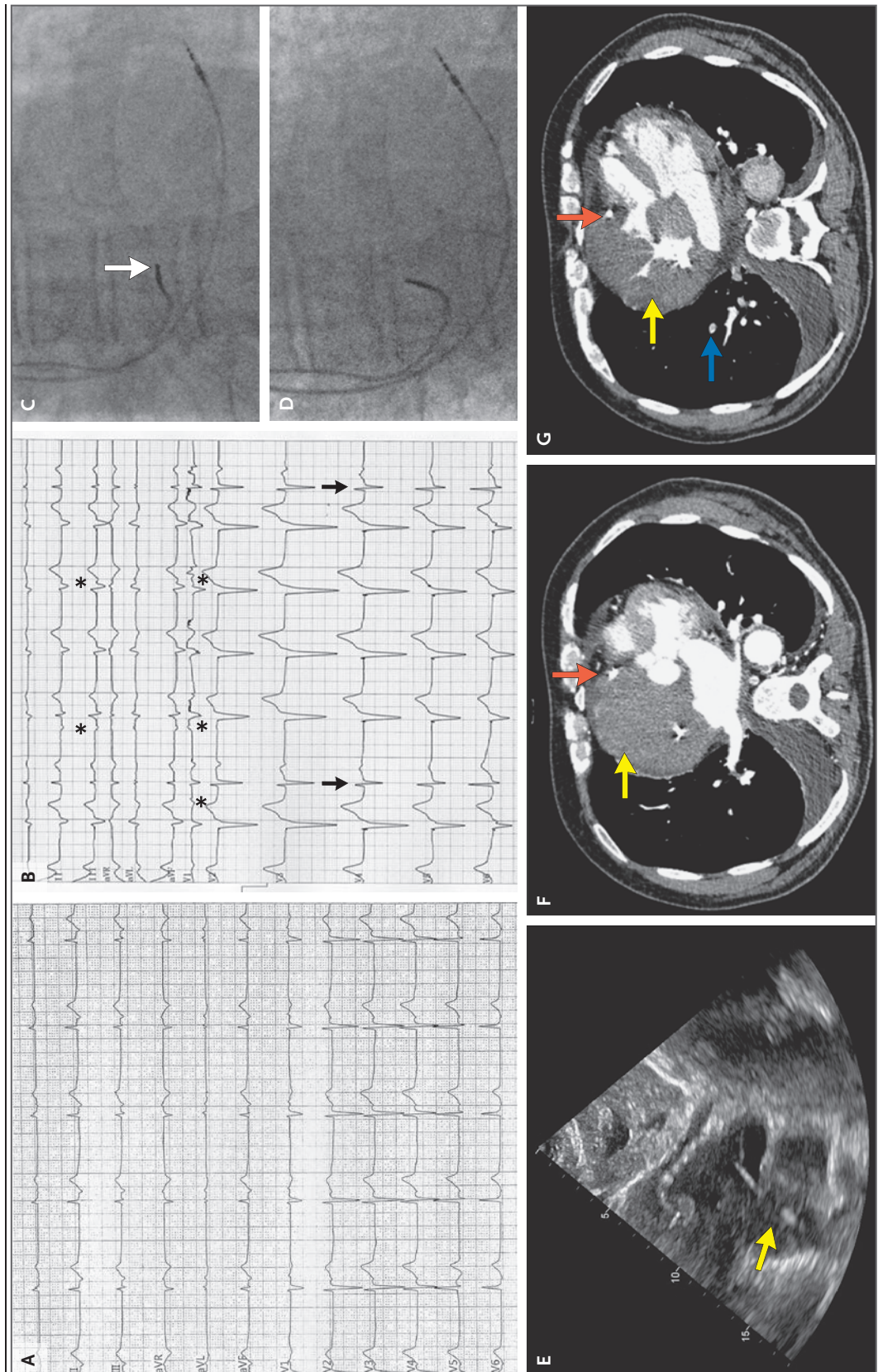


Figure 1 (facing page). Electrocardiograms and Initial Cardiac Imaging.

An electrocardiogram obtained 2 months before the current admission (Panel A) shows sinus arrest with a junctional escape rhythm at rate of 46 beats per minute; retrograde P waves are present, as indicated by the regular negative deflections in the ST segment and T wave after the QRS complex. An electrocardiogram obtained on admission (Panel B) shows ventricular pacing and evidence of atrioventricular dissociation (by comparison, regular atrial activity, which is best seen in leads III and V₁, is indicated by asterisks) and capture beats (arrows). Fluoroscopy performed on admission (Panel C) shows that the location of the atrial pacemaker lead (arrow) appears to have changed from that seen on fluoroscopy that had been performed at the time of lead placement (Panel D), despite both images showing a similar imaging plane. A transthoracic echocardiogram obtained in the subcostal view (Panel E) shows a 5-cm mass (arrow) that obliterates the right atrium and encompasses the pacemaker lead. Axial images from CT angiography of the chest, performed after the administration of intravenous contrast material (Panels F and G), show a nonenhancing mass (yellow arrows) that obliterates the right atrium. Panel F shows a more superior axial slice, obtained at the level of the aortic root, whereas Panel G shows a more inferior axial slice, obtained at the level of the coronary sinus. The presence of the floating-vessel sign (red arrows) suggests that the mass encases coronary vasculature. Pleural effusions and subsegmental pulmonary emboli (Panel G, blue arrow) are also present.

After pacemaker implantation, the patient's exertional symptoms resolved. Treatment with metoprolol was restarted, and he was discharged home.

One month after discharge, dyspnea on exertion developed, and the patient had swelling of the face, arms, and legs. Over the course of the next 4 weeks, dyspnea persisted and edema worsened. The patient was evaluated in his outpatient cardiologist's office and, from there, was admitted to the Third Affiliated Hospital of Sun Yat-sen University.

At the hospital, the patient reported no weight change, anorexia, dysphagia, odynophagia, voice change, ocular symptoms, chest pain or pressure, orthopnea, paroxysmal nocturnal dyspnea, hemoptysis, cough, palpitations, dizziness, light-headedness, syncope, new rashes, gastrointestinal symptoms, or neurologic symptoms. Medical history included infection with severe acute respiratory syndrome coronavirus 2 approximately 6 months earlier. Medications included aspirin, clopidogrel, atorvastatin, metoprolol, and perindopril. He had no known adverse reactions to

medications. There was no known family history of cardiovascular disease or cancer. The patient was a retired physician and lived in an urban area of southern China with his spouse. He had smoked a half pack of cigarettes daily for 40 years but had quit approximately 10 years before the current admission. He did not drink alcohol or use illicit drugs.

On examination, the axillary temperature was 36.1°C, the heart rate 60 beats per minute, the blood pressure 122/70 mm Hg, the respiratory rate 16 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The face appeared edematous without discoloration. The jugular venous pressure was 15 cm of water. Auscultation of the chest revealed bibasilar rales. There was 3+ symmetric edema in the arms and legs. The remainder of the examination was normal.

The white-cell count and differential count and the levels of bilirubin, alkaline phosphatase, glycated hemoglobin, thyrotropin, and fibrinogen were all normal, as were the prothrombin time and partial-thromboplastin time; other laboratory test results are shown in Table 1. A test for human immunodeficiency virus (HIV) types 1 and 2 was negative. In accordance with the standard protocol used at the Third Affiliated Hospital of Sun Yat-sen University, the patient underwent laboratory testing that is routinely performed in older patients on admission, including testing for blood levels of alpha-fetoprotein, CA 125, CA 19-9, and carcinoembryonic antigen; all test results were reported to be normal. A blood sample was obtained for culture.

An electrocardiogram (Fig. 1B) showed ventricular pacing and evidence of atrioventricular dissociation. Pacemaker interrogation showed that the pacing and sensing thresholds of the right ventricular lead were normal, but the pacing threshold of the right atrial lead was greater than 8.0 V, the sensing threshold was undetectable, and the pacing impedance was 1197 ohms. Fluoroscopy (Fig. 1C and 1D) showed that the position of the atrial lead had changed from that seen at the time of lead placement 2 months earlier.

Dr. Xujing Xie: Transthoracic echocardiography (Fig. 1E) revealed an echodensity in the right atrium that measured 5 cm by 4 cm with heterogeneous and hypoechoic internal areas; the mass had not been present on transthoracic echocardiography

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, Third Affiliated Hospital of Sun Yat-sen University†	On Admission
Hemoglobin (g/liter)	130–175	101
Hematocrit (%)	40–50	32
Red-cell count (per μ l)	4,300,000–5,800,000	3,610,000
White-cell count (per μ l)	3500–9500	5240
Platelet count (per μ l)	100,000–350,000	165,000
Sodium (mmol/liter)	137–147	139
Potassium (mmol/liter)	3.50–5.30	4.29
Chloride (mmol/liter)	99–110	103
Carbon dioxide (mmol/liter)	20.2–29.2	22.1
Urea nitrogen (mmol/liter)	2.40–8.20	8.96
Creatinine (μ mol/liter)	31.8–116.0	123.0
Glucose (mmol/liter)	3.90–6.10	8.26
Calcium (mmol/liter)	2.03–2.65	2.31
Aspartate aminotransferase (U/liter)	15–40	43
Alanine aminotransferase (U/liter)	3–35	45
High-sensitivity troponin I (ng/ml)	0.000–0.023	0.012
N-terminal pro-B-type natriuretic peptide (pg/ml)	300–1800	1760
D-dimer (ng/ml)	<500	1960
C-reactive protein (mg/liter)	0.0–6.0	15.4

* To convert the values for urea nitrogen to milligrams per deciliter, divide by 0.357. To convert the values for creatinine to milligrams per deciliter, divide by 88.4. To convert the values for glucose to milligrams per deciliter, divide by 0.05551. To convert the values for calcium to milligrams per deciliter, divide by 0.250.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at the Third Affiliated Hospital of Sun Yat-sen University are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

that had been performed 2 months earlier. The mass obliterated the right atrium and encompassed the pacemaker lead. There was evidence of inferolateral regional hypokinesis with global normal ventricular function. Mild tricuspid regurgitation, a mildly elevated estimated right ventricular systolic pressure, and moderate pericardial effusion were noted. Neither the transtricuspid pressure gradient nor the degree of respirophasic variation in intracardiac velocity could be measured. The inferior vena cava and superior vena cava were difficult to see.

Dr. Jiafu Ou: The next day, CT angiography of the chest (Fig. 1F and 1G) was performed after the administration of intravenous contrast material. An irregular mass in the right atrium that measured 9.1 cm by 7.6 cm by 10.4 cm was identified. The floating-vessel sign was present, which indicated that a coronary artery was encased by

the mass; however, there was no vascular involvement, invasion, or compression associated with the mass. Additional findings included multiple enlarged lymph nodes in the mediastinum, coronary-artery calcification, segmental pulmonary emboli in the right upper lobe and right lower lobe, pulmonary nodules in both lungs, moderate pericardial effusion, a dilated inferior vena cava without a mass, and bilateral pleural effusions.

Dr. Huang: Low-molecular-weight heparin was administered subcutaneously, and treatment with oral furosemide, spironolactone, and empagliflozin was started. The pacemaker setting was changed owing to malfunction of the atrial lead. Empirical treatment with cefoperazone in combination with sulbactam was administered intravenously. During the next several days, edema continued to increase and dyspnea persisted.

Dr. David M. Dudzinski: On the sixth hospital day, transesophageal echocardiography (Fig. 2A, 2B, and 2C) was performed; the mass measured 8.4 cm by 7.1 cm by 6.0 cm and obliterated the right atrium. The mass abutted the aortic root and appeared to infiltrate adjacent tissues. It showed varied internal color Doppler signals, a finding that was possibly consistent with entrapped vessels.

Dr. Huang: Blood cultures obtained at the time of the current admission showed no growth.

Diagnostic tests were performed, and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Jinlai Liu: I participated in the care of this patient and am aware of the final diagnosis. This 79-year-old man, a former smoker with known coronary artery disease complicated by a myocardial infarction that had led to stent placement, presented with dyspnea and edema 2 months after placement of a permanent pacemaker because of sinus arrest. The combination of dyspnea and edema in this patient was initially suggestive of severe tricuspid regurgitation after pacemaker implantation, heart failure as a

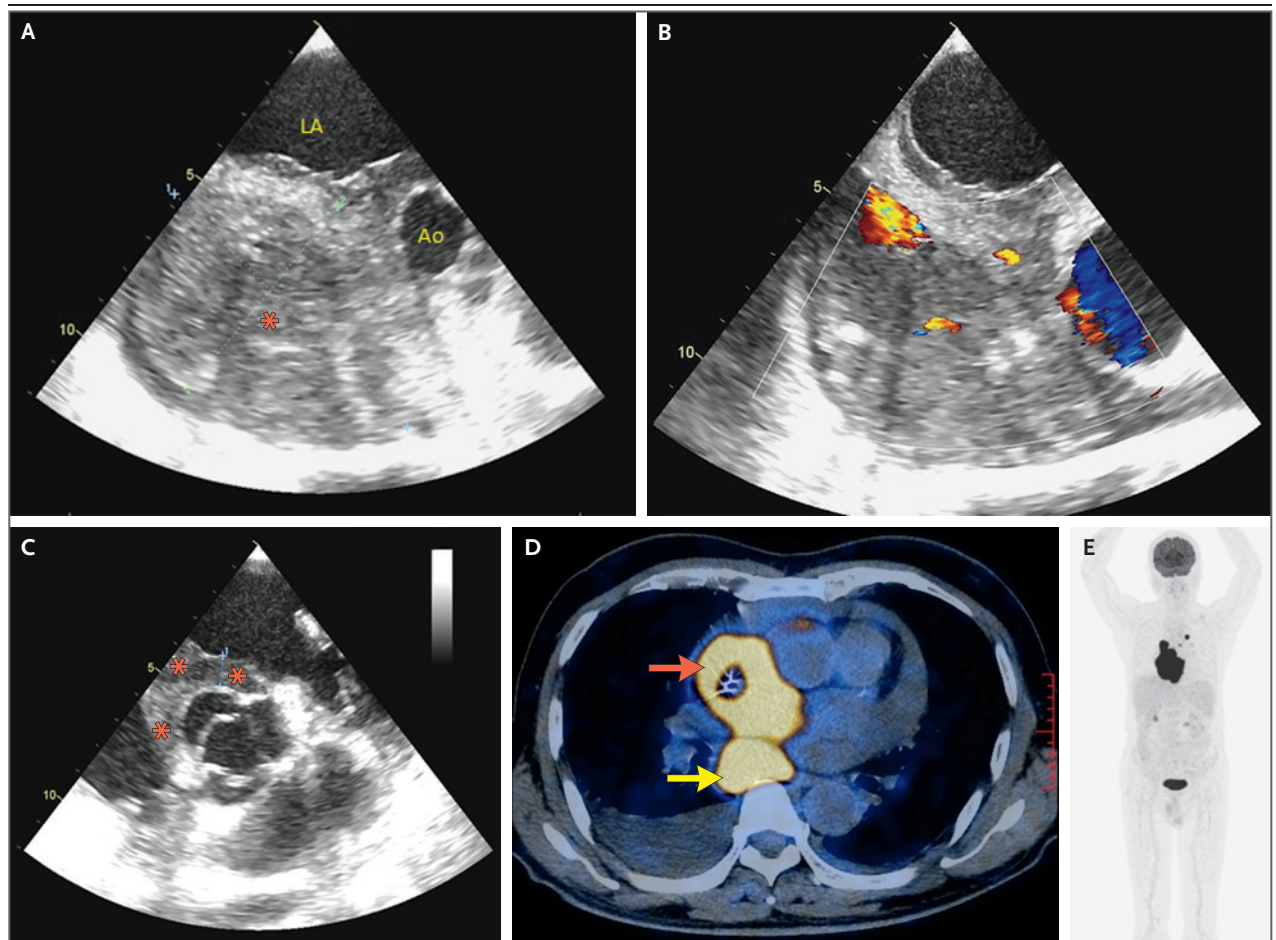


Figure 2. Additional Imaging of the Right Atrial Mass.

A transesophageal echocardiogram, obtained from a midesophageal window after the patient had received 5 days of anticoagulant therapy (Panels A, B, and C), shows that the mass in the right atrium measures 8.4 cm by 7.1 cm by 6.0 cm and completely obliterates the right atrium (Panel A, asterisk); there are varied internal color Doppler signals within the mass (Panel B). The mass also abuts the aortic root (Ao) (Panels A and C) and appears to infiltrate adjacent tissues (Panel C, asterisks). Positron-emission tomographic–CT images (Panels D and E) show ^{18}F -fluorodeoxyglucose (FDG) avidity in the right atrial mass (red arrow). Multiple enlarged and FDG-avid lymph nodes are present in the mediastinum, with the most prominent being an enlarged hilar lymph node (station 10R, yellow arrow). This node would be accessible for diagnostic biopsy by means of endobronchial ultrasound–guided transbronchial needle aspiration. Panel E shows that there are no unexpected FDG-avid locations elsewhere in the body. LA denotes left atrium.

consequence of ischemic cardiomyopathy, superior vena cava syndrome, pericardial effusion, or pericardial constriction. However, initial imaging with transthoracic echocardiography identified a mass in the right atrium, which was considered to be the most likely cause of his symptoms.

The correct identification of the cause of a cardiac mass is a critical first step to facilitate appropriate management and therapy. In working toward the most likely diagnosis in this patient's case, I will consider his age and coexisting conditions, as well as the imaging characteristics of the intracardiac mass on transthoracic echocardiography, CT, and transesophageal echocardiography.¹⁻³

RIGHT ATRIAL THROMBUS

Thrombosis is the most common cause of an intracardiac mass.⁴ A right atrial thrombus is often related to indwelling devices, catheters, or pacemaker leads and may result in pulmonary emboli. In addition, a thrombus may form in the presence of a right atrial tumor. Given this patient's history of pacemaker-lead implantation, pacemaker-lead displacement, and pulmonary emboli seen on CT angiography, a thrombus was initially considered, and he received empirical anticoagulant therapy. However, the rapid enlargement of the mass over the course of 5 days between initial transthoracic echocardiography and subsequent transesophageal echocardiography, despite the use of anticoagulant therapy, made the presence of a thrombus unlikely.

INFECTIVE ENDOCARDITIS

Infective endocarditis involving a vegetation on the right atrial pacemaker lead with emboli to the lungs could explain the findings of a right atrial mass and pulmonary emboli. However, patients with infective endocarditis typically present with signs and symptoms of systemic infection, such as fever, chills, anorexia, and weight loss — none of which this patient had. Although systemic infection may be subtle depending on the causative organism, several aspects of this patient's presentation make infectious endocarditis unlikely. First, the mass would be unusually large for a vegetation. Second, the rapid growth observed while the patient was in the hospital receiving empirical treatment with broad-spec-

trum antimicrobial activity would be unexpected. Third, blood cultures ultimately showed no growth, specifically of organisms traditionally classified in the HACEK group (haemophilus species, *Aggregatibacter actinomycetemcomitans*, *Cardiobacterium hominis*, *Eikenella corrodens*, and *Kingella kingae*).

CARDIAC METASTASES

The incidence of cardiac metastases is 20 to 40 times that of primary cardiac tumors.⁵ Among cancers that metastasize to the heart, the most common is lung cancer, followed by breast cancer, kidney cancer, melanoma, and lymphoma.⁶ This patient had pericardial effusion, which often occurs in patients with cardiac metastasis, and his smoking history increases the risk of multiple types of cancer, especially lung cancer. Although chest CT showed mediastinal lymphadenopathy and lung nodules, none of the masses suggested the presence of lung cancer. Renal-cell carcinoma, hepatocellular carcinoma, and primary leiomyosarcoma of the inferior vena cava can extend directly from the inferior vena cava into the right atrium, manifesting as an intraluminal defect. Although the inferior vena cava was difficult to see on transthoracic echocardiography, there was no evidence of a mass in the inferior vena cava on chest CT. No obvious primary cancer was detected in this patient; however, cardiac metastasis remains a possible diagnosis.

BENIGN PRIMARY CARDIAC TUMORS

Atrial myxomas are the most common benign primary cardiac tumors in adults. Myxomas occur more commonly in the left atrium than the right atrium and are often attached to the interatrial septum, particularly near the fossa ovalis. Myxomas appear on echocardiography as mobile and smooth, with homogeneous echotexture. The mass seen in this patient had an uncertain attachment point, did not appear to be mobile, and had heterogeneous echotexture. These findings, along with the rapid increase in size, make a myxoma unlikely.

Lipomas also have a homogeneous appearance on echocardiography, often occur in the myocardium, and do not typically obliterate the atrial lumen. The rapid growth of the mass in this patient is also inconsistent with a lipoma.

Papillary fibroelastomas, which are another type of benign primary cardiac tumor, usually have a valvular attachment. Rhabdomyomas are the most prevalent benign primary cardiac tumors in infants and children but rarely occur in adults and thus would be unlikely in this patient.⁷

Most cardiac paragangliomas are benign, but malignant features are present in up to 10% of cases.⁸ Cardiac paragangliomas occur most commonly in the left atrium, followed by the interatrial septum; they may occasionally result in encasement of a coronary artery, a feature seen on transthoracic echocardiography in this patient.⁸ Secretory cardiac paragangliomas can cause tachycardia, tremors, palpitations, flushing, hypertension, or hypotension due to secretion of catecholamines such as norepinephrine, epinephrine, or dopamine. Nonsecretory cardiac paragangliomas may produce compressive, embolic, or obstructive symptoms, which were seen in this patient. However, the rapid growth of the mass in this patient makes cardiac paraganglioma unlikely.

MALIGNANT PRIMARY CARDIAC TUMORS

The rapid growth of the atrial mass that was noted on serial imaging in this patient was suggestive of a malignant primary cardiac tumor, although this diagnosis is rare. Approximately 95% of malignant primary cardiac tumors are sarcomas — usually angiosarcomas or rhabdomyosarcomas. Angiosarcomas typically occur in the right atrioventricular groove, often invading the pericardium and right atrial wall.⁹ Areas of necrosis and hemorrhage within angiosarcomas create a heterogeneous appearance on imaging, as was seen in this patient. Pericardial and pleural effusions may also occur with angiosarcomas, since angiosarcomas may exhibit direct pericardial and pleural invasion. Cardiac rhabdomyosarcomas are rare, aggressive, malignant tumors of the heart that are most commonly seen in children but can occur in adults and are associated with a poor prognosis. They can originate in various locations, most frequently in the right atrium, followed by the left atrium and ventricles.

The second most common malignant primary cardiac tumor is primary cardiac lymphoma, the incidence of which is increasing worldwide. Risk factors include infection with HIV,¹⁰ infec-

tion with Epstein–Barr virus, and organ transplantation. The most common histologic type of primary cardiac lymphoma is diffuse large B-cell lymphoma (DLBCL), followed by follicular B-cell lymphoma, Burkitt's lymphoma, T-cell lymphoma, and others.^{2,5} Primary cardiac lymphomas have a predilection for the right side of the heart, including the right atrium. Persons with primary cardiac lymphoma may present with protean constitutional symptoms, dyspnea, rhythm disturbances and heart blocks, and heart failure. On echocardiography, primary cardiac lymphomas are associated with a variety of findings, including wall thickening, infiltrative masses, or nodular masses. On CT of the chest, concurrent mediastinal lymphadenopathy and pulmonary nodules may also be present. The floating-vessel sign that was seen on CT angiography of the chest in this patient has been reported in patients with aggressive lymphomas that encase the right coronary artery and is highly suggestive of primary cardiac lymphoma.¹¹

When I evaluated this patient, I thought that his presentation was most consistent with a rapidly growing malignant primary cardiac tumor that was causing mass effect in the atrium and affecting the pacemaker lead, with primary cardiac lymphoma being the most likely diagnosis. To establish the diagnosis, I would perform pericardiocentesis with cytologic analysis of pericardial fluid, which has a sensitivity of approximately 60% for the diagnosis of primary cardiac lymphoma. If cytologic examination of the fluid is nondiagnostic, biopsy of the atrial mass could be performed with the use of a percutaneous endovascular approach, given the size and location of the mass. However, risks include bleeding, arrhythmia, perforation of the cardiac chamber, and embolization. Another possible option would be biopsy of involved lymph nodes with the use of a peripheral, mediastinal, or endobronchial approach. Positron-emission tomography (PET)–CT could help assess which node to target.

DR. JINLAI LIU'S DIAGNOSIS

Pacemaker-lead dislodgement due to a malignant primary cardiac tumor, most likely primary cardiac lymphoma.

DIAGNOSTIC TESTING

Dr. Huang: The blood level of lactate dehydrogenase was elevated (845 U per liter; reference value, <250).

Dr. Ou: ^{18}F -fluorodeoxyglucose (FDG) PET-CT (Fig. 2D and 2E) was performed to help identify a primary cancer and provide information about possible sites for biopsy.¹² The images showed marked FDG avidity overlying the right atrial mass. Adjacent intense FDG uptake was noted in a mediastinal lymph node at the right hilum (station 10R), and multiple other metabolically active mediastinal nodes were detected. This pattern of lymph-node involvement suggested lymphoma — most likely primary cardiac lymphoma rather than a metastatic lymphoma, given that no other sites of FDG avidity were found.

Dr. Chunkui Shao: Pericardiocentesis was performed, and 300 ml of bloody fluid was drained. Cytologic analysis of the pericardial fluid showed numerous lymphocytes, with a small number of neutrophils and mesothelial cells (Fig. 3A); no clearly abnormal-appearing cells were observed.

Dr. Liu: Because the analysis of the pericardial fluid was nondiagnostic, a tissue biopsy was required for diagnosis. Transvenous biopsy of the right atrial mass was considered to confer a high risk because of the possibility of distal embolization or disruption of the encased vessel or aorta; there is limited experience at our center with this type of biopsy. After a comprehensive

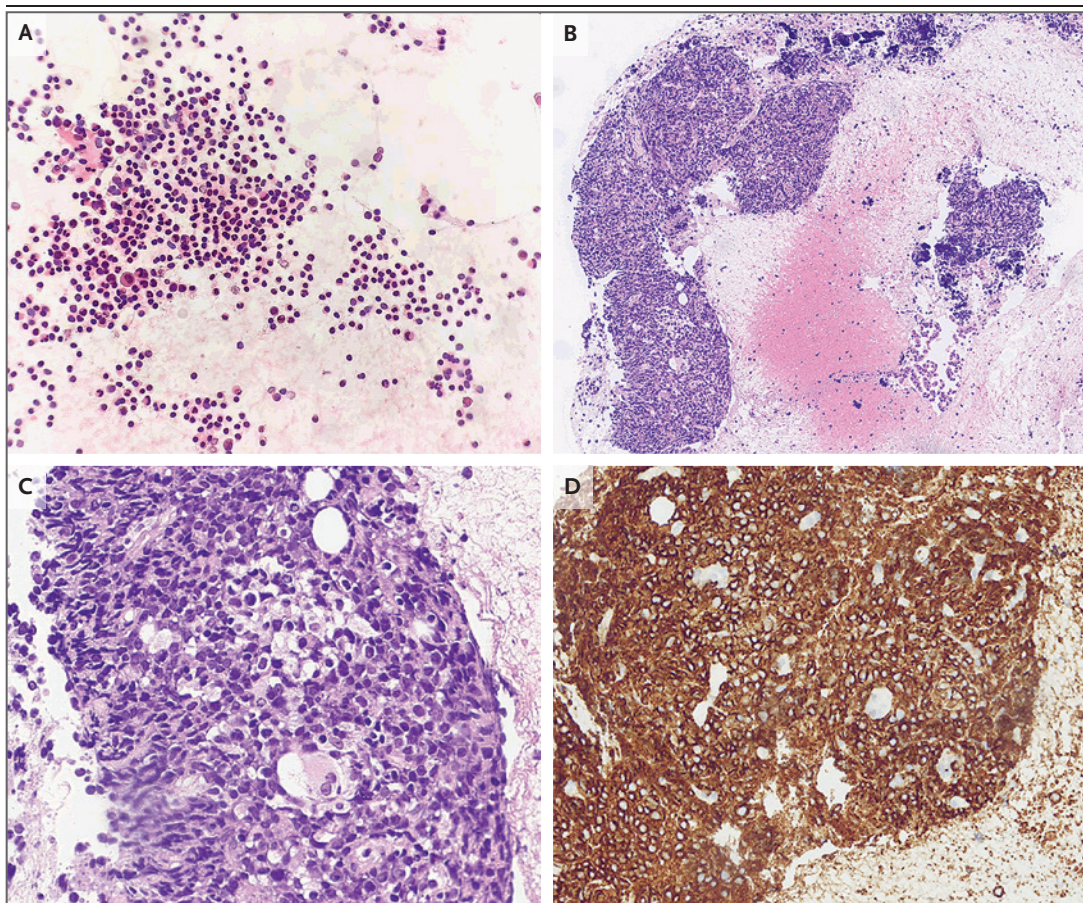


Figure 3. Pericardial Fluid Sample and Biopsy Specimen of a Mediastinal Lymph Node.

Hematoxylin and eosin staining of fluid obtained from the pericardial effusion (Panel A) shows prominent lymphocytes. Hematoxylin and eosin staining of a biopsy specimen of a mediastinal lymph node (station 10R) (Panels B and C [higher magnification]) shows architectural effacement resulting from a diffuse proliferation of medium and large lymphoid cells. Immunohistochemical staining of the biopsy specimen (Panel D) shows that the tumor cells are strongly positive for the B-cell marker CD20.

multidisciplinary discussion, we pursued endobronchial ultrasound–guided transbronchial needle aspiration biopsy of the right hilar lymph node that had shown marked FDG avidity on PET-CT.

PATHOLOGICAL DISCUSSION

Dr. Shao: Hematoxylin and eosin staining of the biopsy specimen obtained from the right hilar lymph node (Fig. 3B and 3C) showed architectural effacement resulting from a diffuse proliferation of medium and large lymphoid cells. The neoplastic cells had a uniform cytologic appearance with prominent single central nucleoli and an appreciable amount of basophilic cytoplasm. Histologic features were consistent with large B-cell lymphoma.

Immunohistochemical staining of the specimen showed that most of the cells strongly expressed B-cell markers such as CD20 (Fig. 3D) and CD79a. Immunohistochemical staining was positive for B-cell lymphoma 6 protein (BCL6) and multiple myeloma oncogene 1 (MUM1). In addition, although staining for CD10 was negative, staining for B-cell lymphoma 2 (BCL2) was moderately to strongly positive in approximately 90% of the tumor cells; staining for C-MYC was moderately to strongly positive in approximately 40% of the tumor cells. The proliferation marker Ki-67 was detected in approximately 95% of the tumor cells, a finding that indicated rapid cell turnover and growth. In situ hybridization for Epstein–Barr virus–encoded small RNA was negative. These histopathological findings confirmed the diagnosis of DLBCL (nongerminal center B-cell subtype). A bone marrow biopsy did not show bone marrow involvement.

PATHOLOGICAL DIAGNOSIS

Primary cardiac lymphoma, specifically diffuse large B-cell lymphoma (nongerminal center B-cell subtype).

DISCUSSION OF MANAGEMENT

Dr. Jingyun Wen: Cardiac lymphoma may occur as a primary neoplasm, wherein the lymphoma originates in the heart, or as a secondary metastatic cancer, wherein the lymphoma spreads to the heart from another part of the body.^{13,14} Primary cardiac lymphoma is rare, accounting for

1.3% of primary cardiac tumors.¹⁵ The most common type of primary cardiac lymphoma is DLBCL. Secondary cardiac lymphoma is more prevalent than primary cardiac lymphoma; cardiac involvement has been identified in up to 25% of autopsies performed in patients with lymphoma.^{13,14}

The Hans algorithm has been developed to determine the cell of origin of DLBCL by means of immunohistochemical staining with the use of antibodies against CD10, BCL6, and MUM1 and to categorize DLBCL into germinal center B-cell and nongerminal center B-cell subtypes (the latter of which has a poorer prognosis).¹⁶ Primary cardiac DLBCL is usually of the nongerminal center B-cell phenotype, and many tested samples of primary cardiac lymphoma express both BCL2 and C-MYC,^{17,18} as was seen in this patient.

Primary cardiac lymphoma progresses rapidly and is often diagnosed at an advanced stage. Untreated patients typically die within 1 month after diagnosis, and with contemporary treatment, the median overall survival is 7 to 12 months. The incidence of recurrence is high. Risk factors for a poor prognosis include advanced age, coexisting extracardiac disease (mediastinal lymphoma), and arrhythmias — all of which were present in this patient — as well as left ventricular involvement associated with primary cardiac lymphoma and immunosuppression.¹⁰

We planned for the patient to receive a regimen of rituximab (an anti-CD20 monoclonal antibody), vincristine, and prednisone as the first cycle of treatment owing to his frailty and concern about possible tumor lysis syndrome as a result of his large tumor burden. After the patient was found to have minimal side effects during the first cycle, the second cycle was based on a standard R-CHOP regimen (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone).

Dr. Ou: Seven weeks after the initiation of chemotherapy, chest CT, performed after the administration of intravenous contrast material (Fig. 4A and 4B), showed a marked reduction in the size of the right atrial mass (4.8 cm by 3.5 cm by 3.1 cm). Mediastinal lymph nodes had also reduced in size.

Dr. Liu: Pacemaker interrogation showed that the pacing threshold of the right atrial lead had decreased to 5.0 V, the sensing threshold had

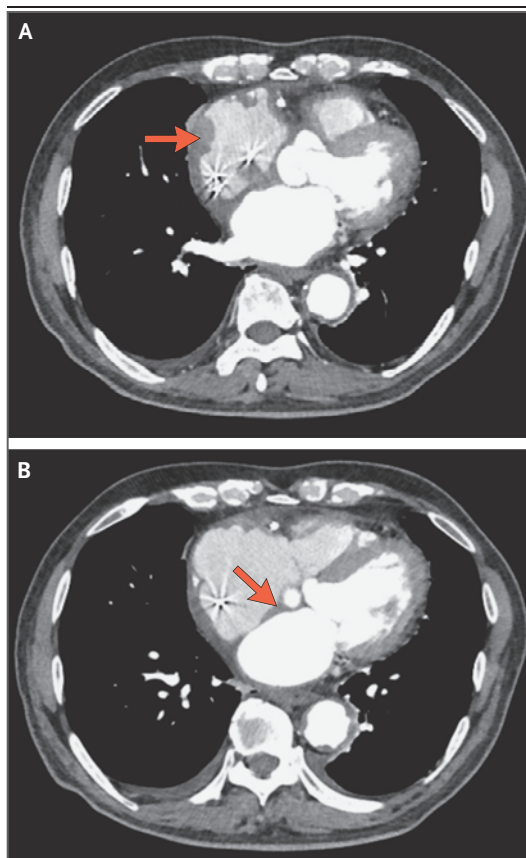


Figure 4. Follow-up Chest CT.

Axial images from follow-up CT of the chest obtained after two cycles of chemotherapy (Panels A and B) show that the size of the right atrial mass is markedly reduced (red arrows) as compared with initial CT images. Panel A shows a more superior axial slice, obtained at the level of the aortic root, whereas Panel B shows a more inferior axial slice, obtained at the level of the coronary sinus.

improved to 1.0 mV, and the pacing impedance had normalized to 627 ohms. These findings suggested that the connection of the right atrial lead with the atrial myocardium had improved. We attributed this improvement to the administration of chemotherapy; after treatment, a reduction in the size of the mass was noted, along with a possible decrease in myocardial infiltration by the primary cardiac lymphoma, with reduced displacement of the atrial lead.

FOLLOW-UP

Dr. Wen: Within 2 months after chemotherapy was initiated, edema resolved, and the patient

was able to perform activities of daily living without dyspnea and with only mild chest tightness after physical activity. The N-terminal pro-B-type natriuretic peptide level had decreased to 746 pg per milliliter. The blood level of lactate dehydrogenase had decreased to 190 U per liter.

Seven weeks after this follow-up chest CT, leg edema recurred, and dyspnea and new fatigue developed, despite treatment with three additional cycles of R-CHOP. Repeat chest CT was performed and showed that the mass had increased in size to 7.0 cm by 7.0 cm by 4.3 cm. In addition, the blood level of lactate dehydrogenase had increased to 527 U per liter. Owing to disease progression, the chemotherapy regimen was switched to rituximab, gemcitabine, and oxaliplatin; however, no clinical improvement was observed after three cycles during the subsequent 8 weeks. The blood level of lactate dehydrogenase had further increased to 1007 U per liter. Subsequent cycles of a regimen of rituximab, lenalidomide, and the Bruton's tyrosine kinase inhibitor orelabrutinib were administered, which caused facial rash.

Other approaches for treatment of primary cardiac lymphoma could include radiation therapy, autologous stem-cell transplantation, and chimeric antigen receptor T-cell therapy (also known as CAR T-cell therapy), but for this patient of advanced age, the chemotherapy regimen was switched to rituximab, methotrexate, and vincristine. Unfortunately, despite these changes to his treatment plan, the patient's disease continued to progress and he died 10 months after his initial diagnosis.

Dr. Xie: Could the sinus arrest in this patient have been related to the diagnosis of primary cardiac lymphoma?

Dr. Liu: The highest suspicion at that time focused on myocardial ischemia or ischemic damage to the conduction system, but the coronary angiography findings were unchanged from those of the previous study obtained after percutaneous coronary intervention, and other evaluations for ischemia were unrevealing. We theorize that occult myocardial infiltration associated with primary cardiac lymphoma could have contributed to dysfunction of the sinoatrial node. This theory is corroborated by initial difficulty in placing the right atrial lead owing to high pacing thresholds, which in retrospect suggests that lymphoma had invaded the subendocardium

2 months before the overt clinical presentation, despite the fact that the tumor had not yet been detected on imaging.

FINAL DIAGNOSIS

Primary cardiac diffuse large B-cell lymphoma.

This case was presented at the Asian Heart Society International Cardiovascular Diseases Research and Prevention Strategy seminar as part of a Sino-American Expert Grand Rounds conference, held in Guiyang, China, on April 20, 2024, and directed by GanXin Yan, Ruihai Zhou, Tiangang Zhu, Jiafu Ou, and Lanqi Hua.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank all the doctors who participated in the diagnosis and treatment of this patient at the Third Affiliated Hospital of Sun Yat-sen University.

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